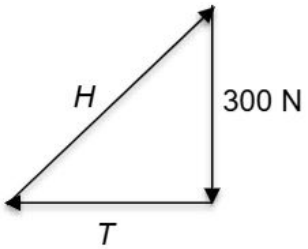
1	(a)	 [M1] for correct vector triangle with correct direction [A1] for correct labels of forces [deduct one mark for label angle 30°]
	(b)	By taking moments about the hinge, $T \times 0.60 = 300 \times 1.5 \cos 30°$ [C1] $T = 650$ N [A1]
	(c)	Horizontally, $\Sigma F_x = 0$ $T = H_x$, $H_x = 650$ N Vertically, $\Sigma F_y = 0$ $H_y = 300$ N [C1 for both equations] $H = \sqrt{H_x^2 + H_y^2} = \sqrt{650^2 + 300^2}$ $= 716$ N [A1] $\theta = \tan^{-1}\left(\frac{H_y}{H_x}\right)$ $= \tan^{-1}\left(\frac{300}{650}\right)$ $= 24.8°$ [A1] 24.8° anticlockwise above the horizontal.